

Lecture12: Tree1

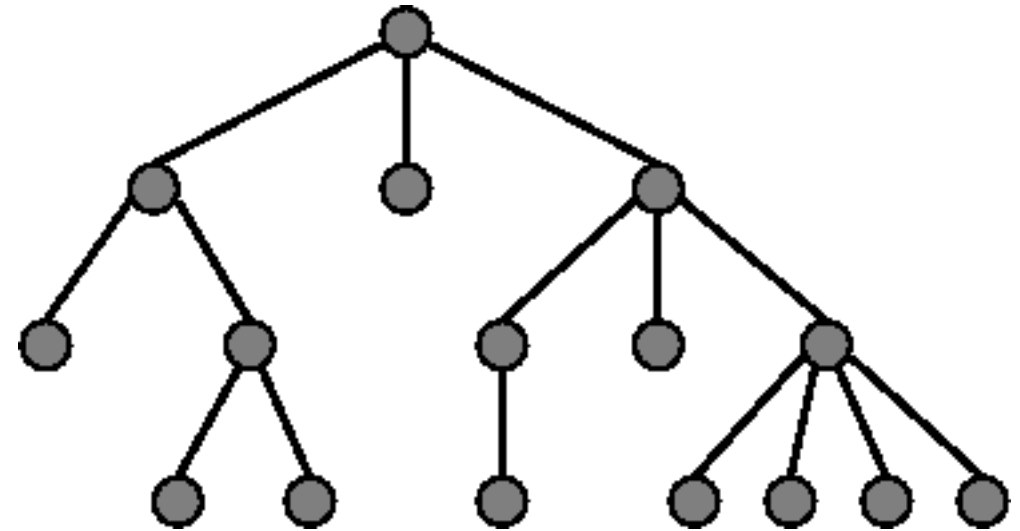
STA 4724

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In This Chapter

- In this chapter we will address a few techniques that have been inspired by the representation of a tree.
- root, branch, leaves, nodes, parent, children

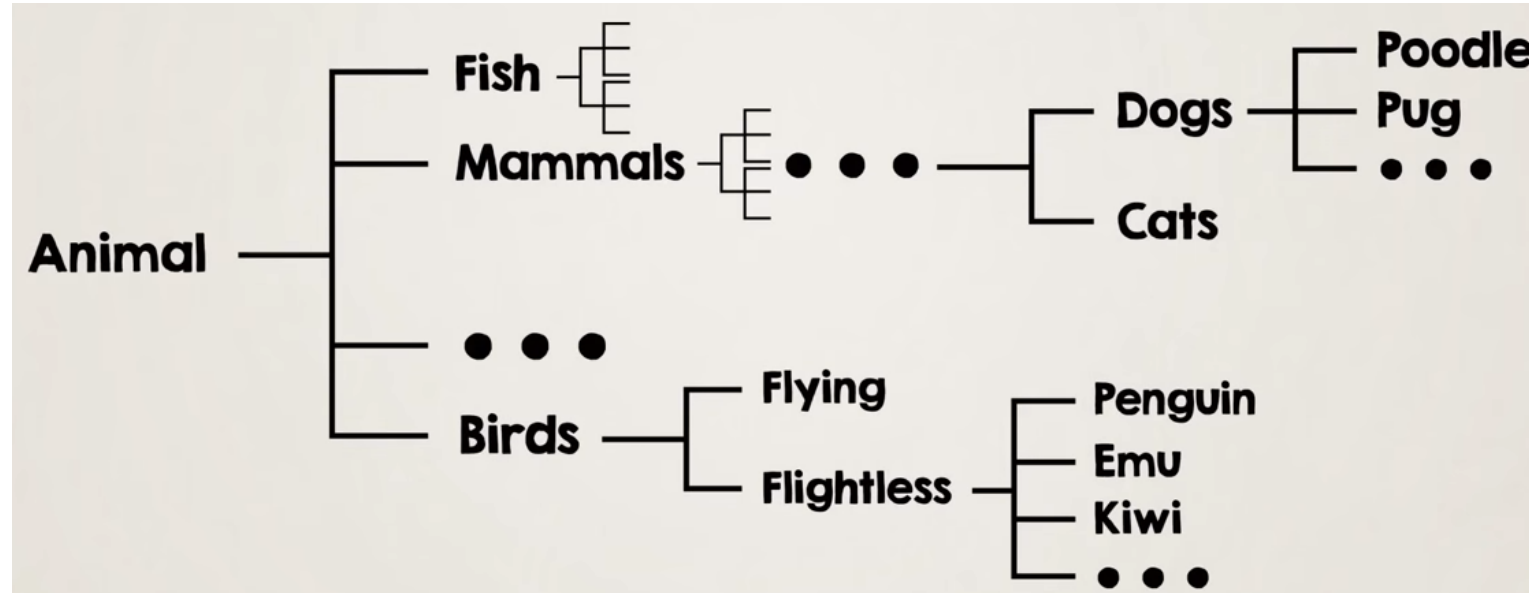


Hierarchical Clustering

- There are two types of clustering: flat and hierarchical
- K-means is an algorithm of flat clustering
- Informally, Hierarchical Clustering is the Clustering of Clustering
- Store -> Food, electronic, furniture
 - Food -> veggie, meat, seafood
 - Electronic -> TV, computer, console
 - Computer -> laptop, desktop, workstation
 - Laptop -> gaming, office, 2-in-1
- An unsupervised learning task



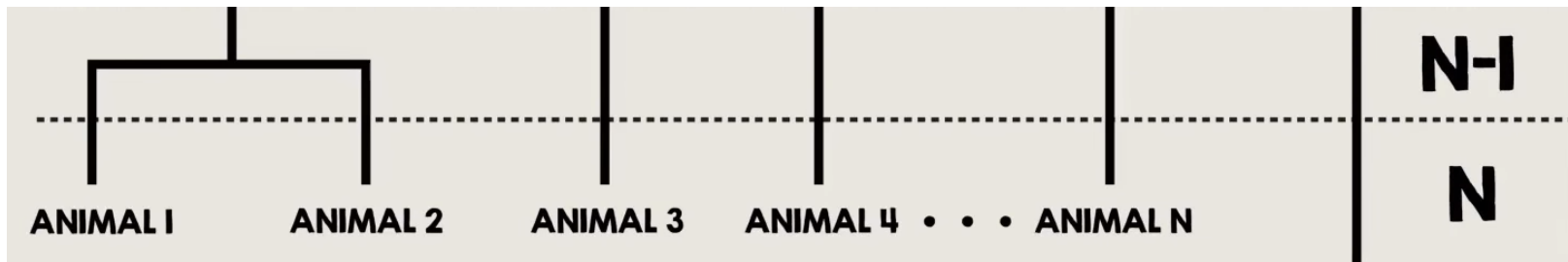
A Tree Structure



- There are two kinds of Hierarchical clustering:
 - Agglomerative (Bottom-up)
 - Divisive (Top-down)

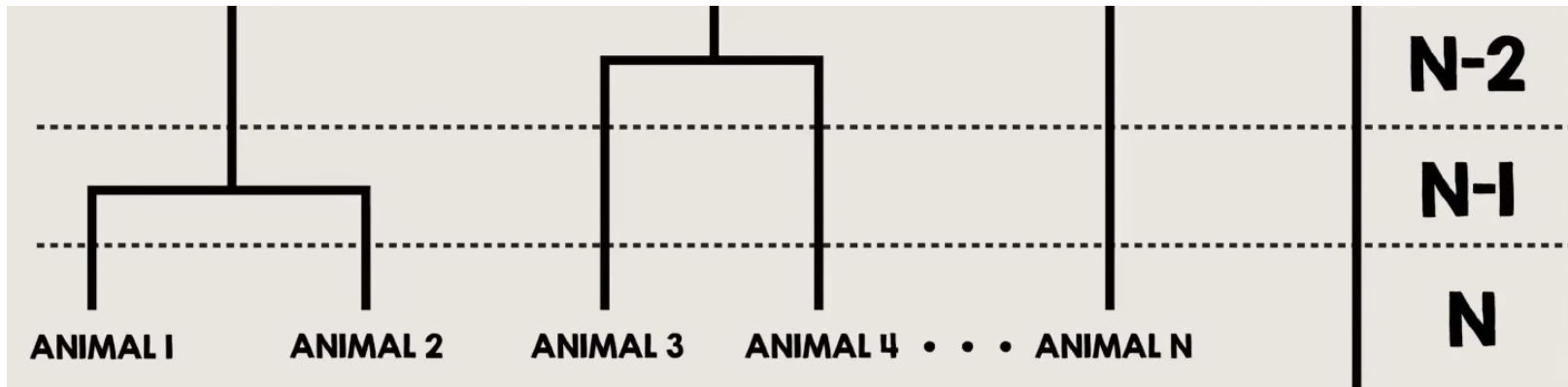
Agglomerative (Bottom-up)

- Find the two most close/similar items (clusters)
 - Use some measurements such as Euclidean distance or cosine angle



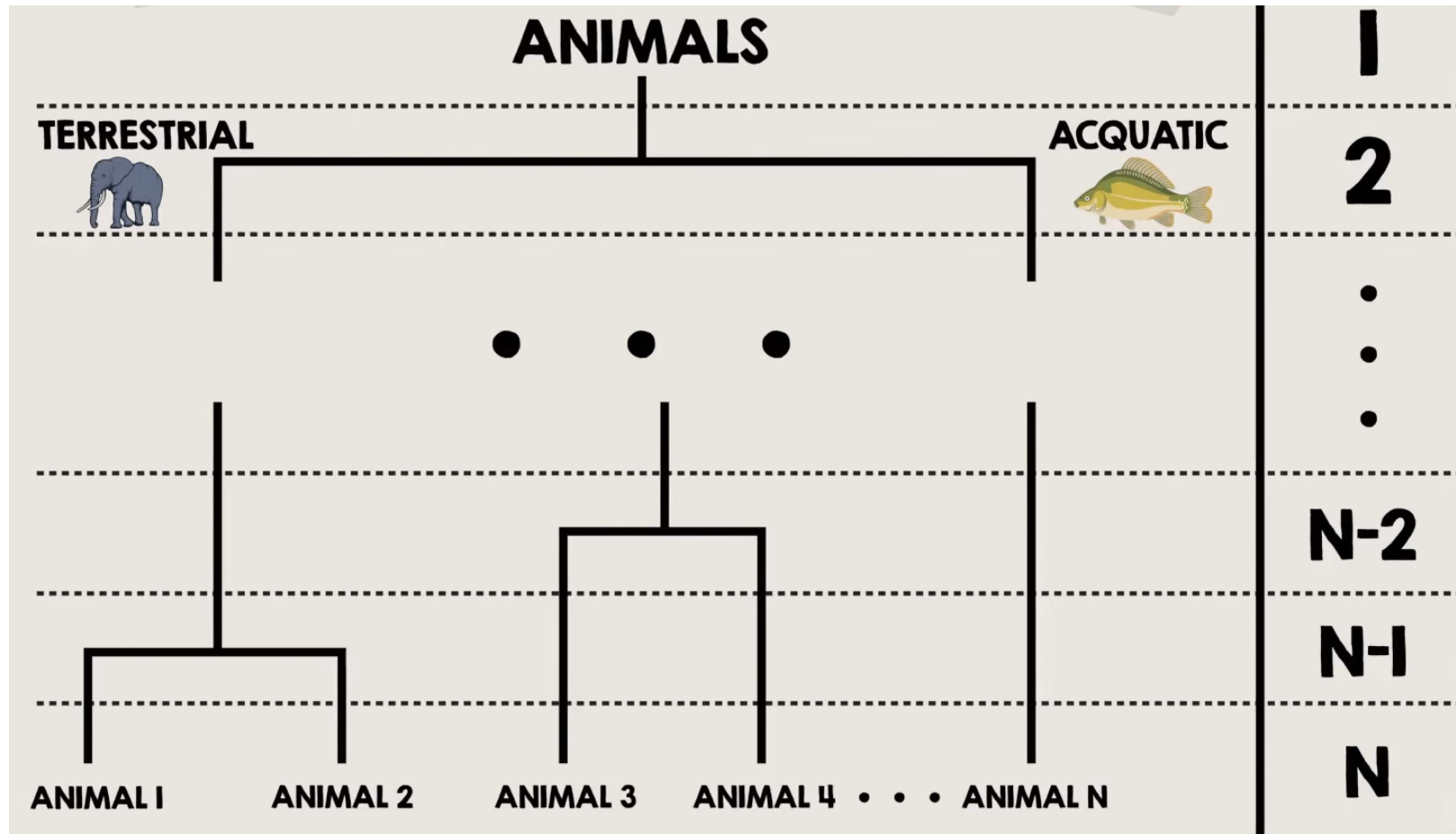
Agglomerative (Bottom-up)

- Find the next two most close items (clusters)
 - Now the cluster of Animals 1 & 2 is viewed as one item (cluster)

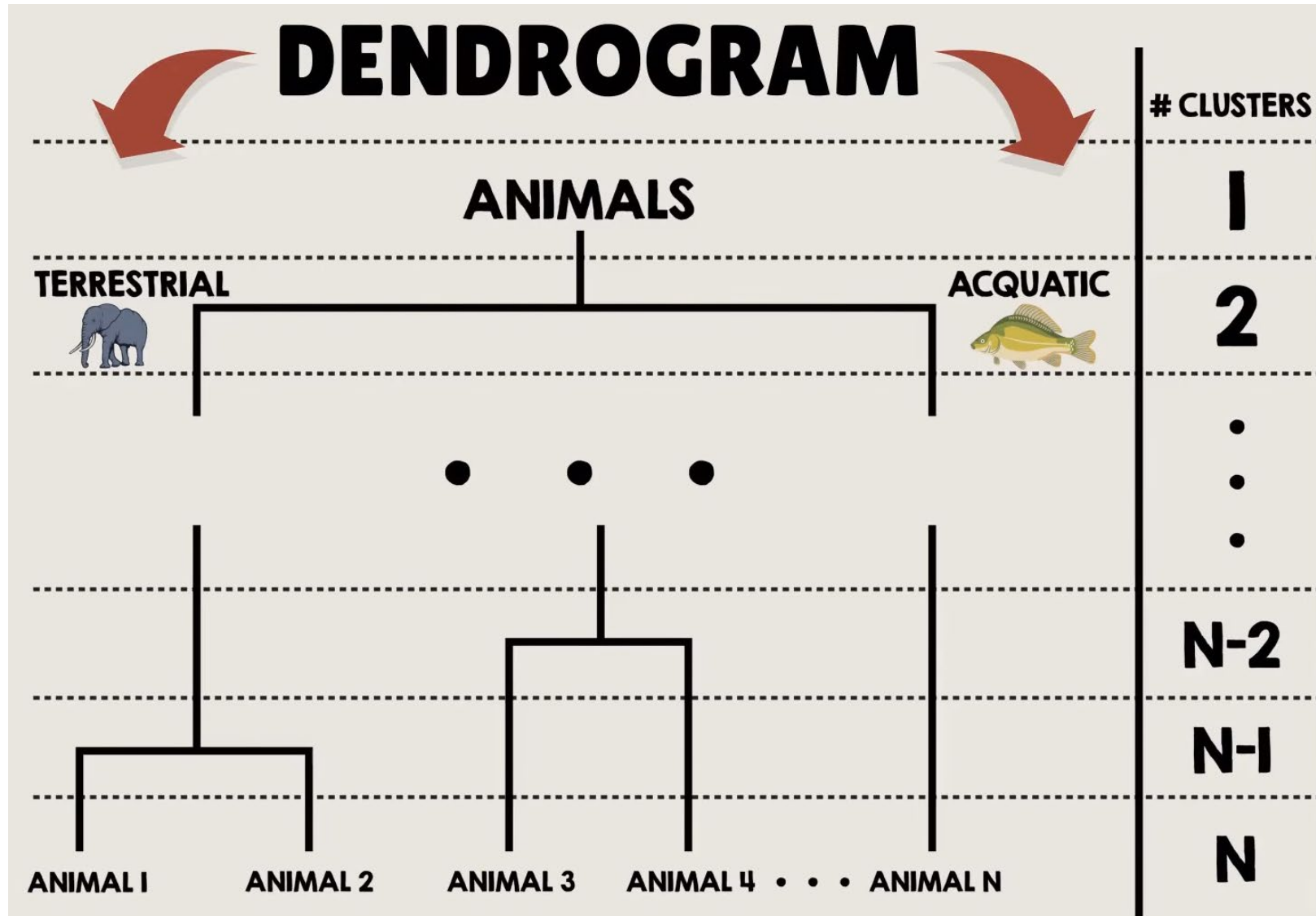


Agglomerative (Bottom-up)

- Repeat these steps until we have only one cluster

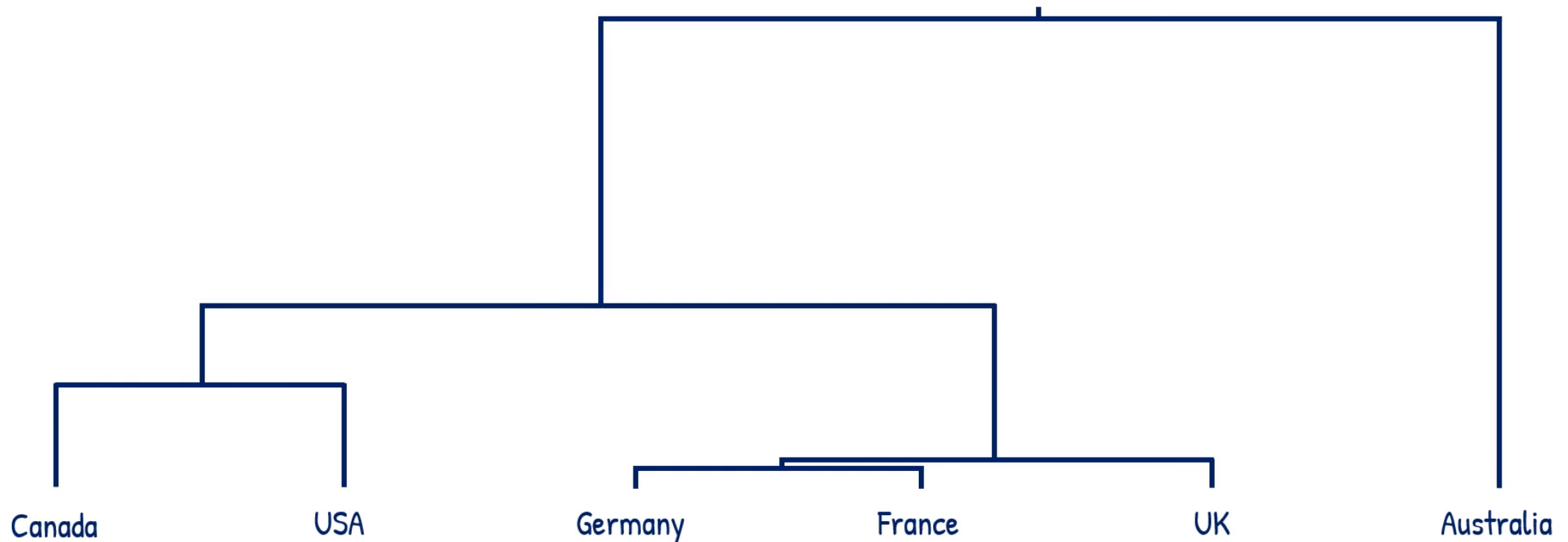


The tree-like structure is called Dendrogram.



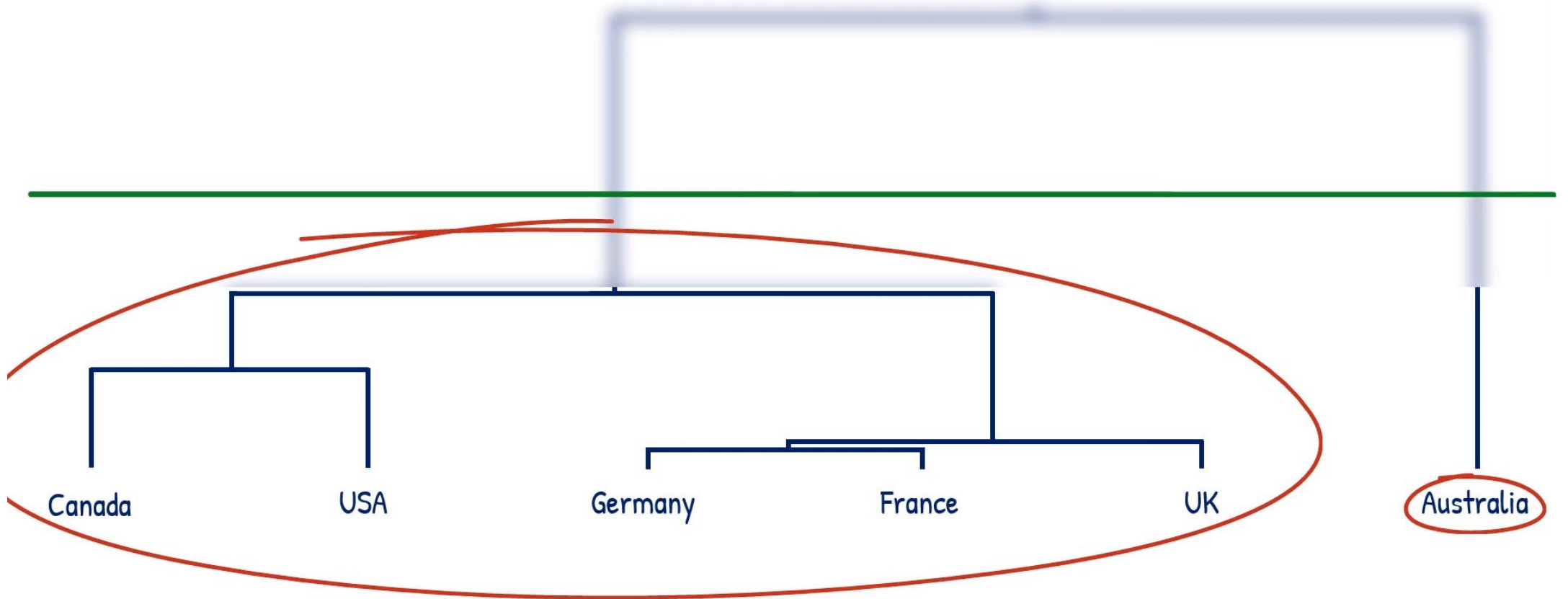
You can choose any K at anytime!

- In dendrogram, the heights of the links represent the similarity between the items
- You don't have to preset K 's value (the number of clusters)
- You can have K -clustering for any K , by drawing a horizontal line



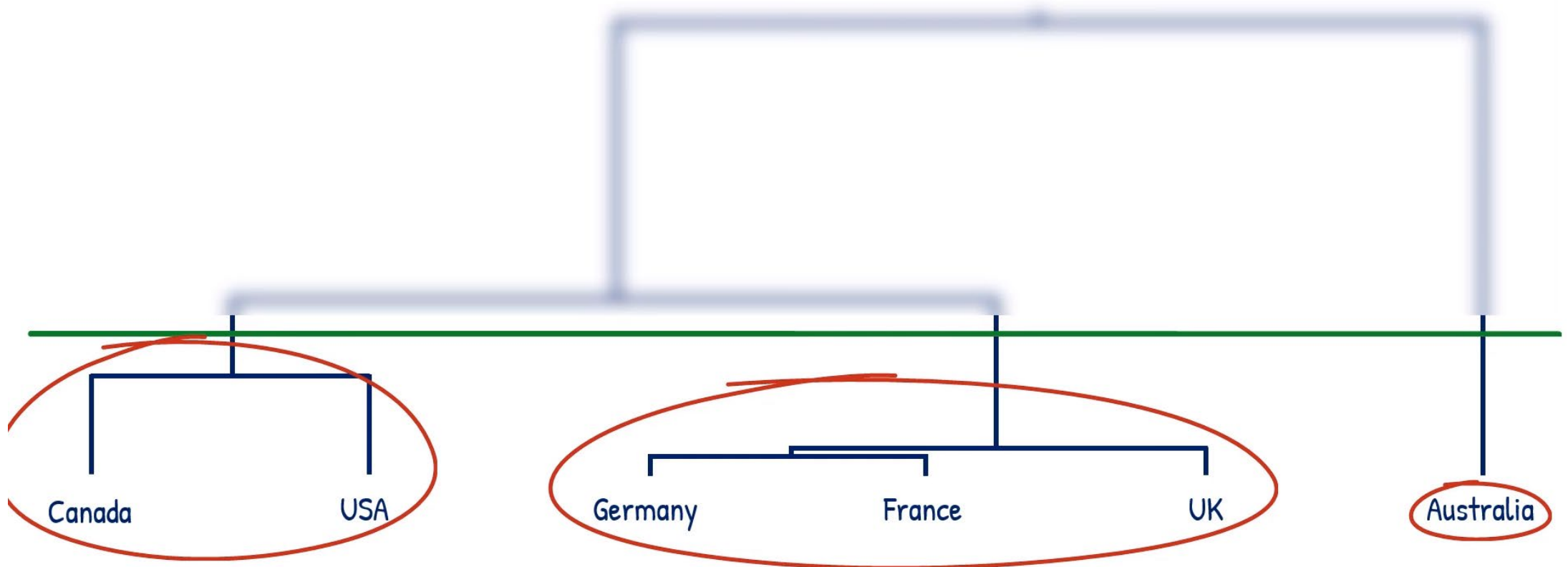
You can choose any K at anytime!

- 2-clustering



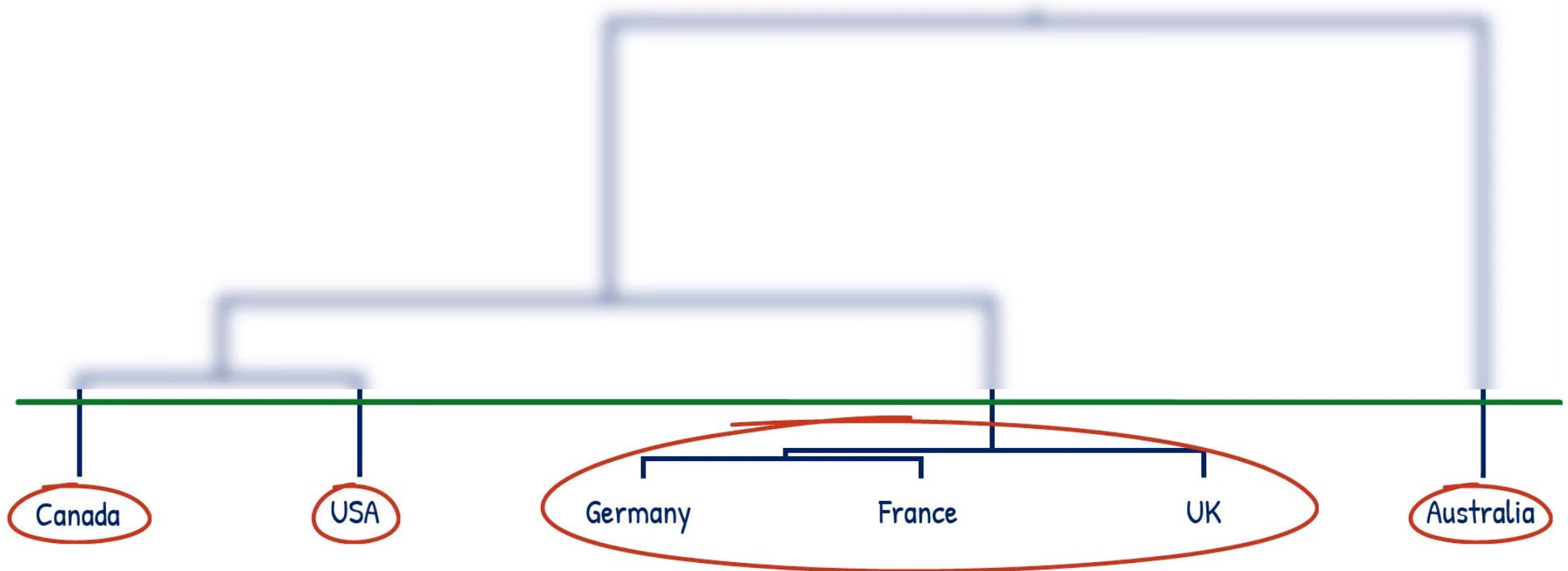
You can choose any K at anytime!

- 3-clustering



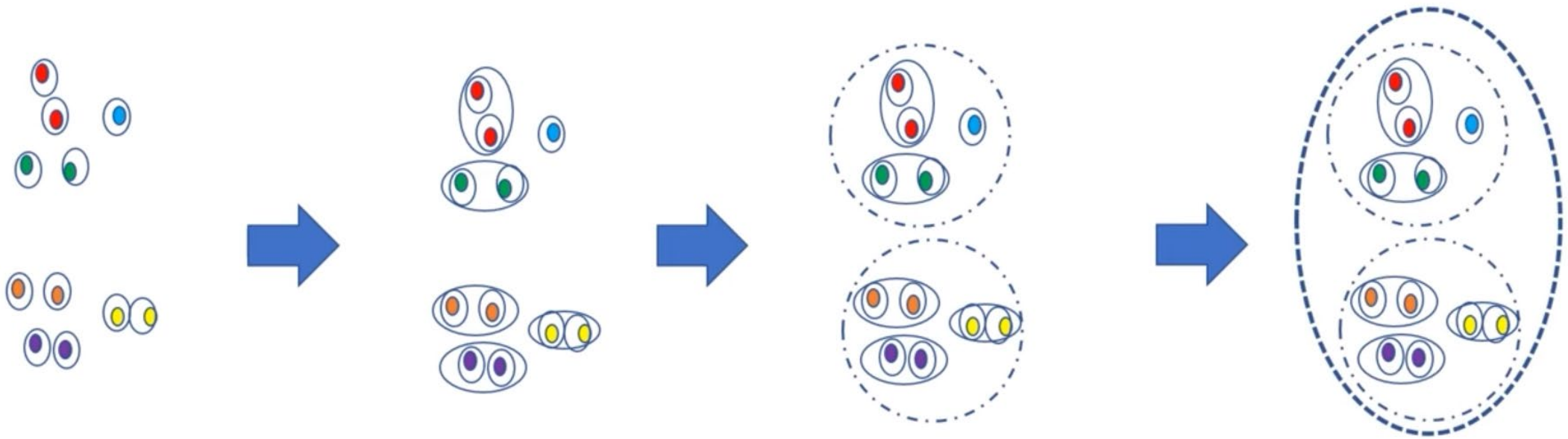
You can choose any K at anytime!

- 4-clustering



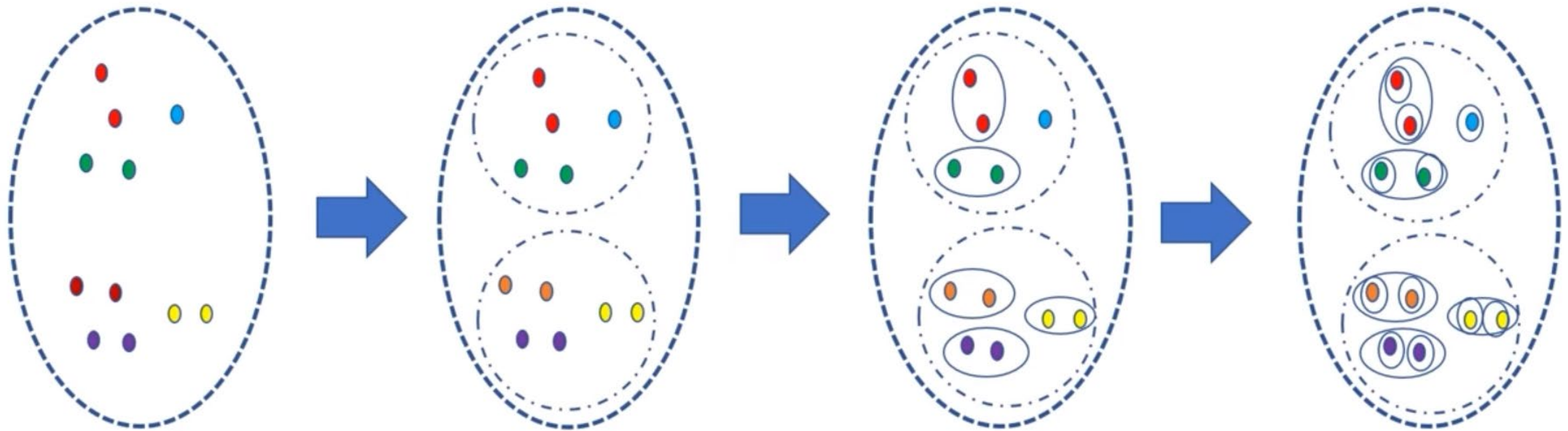
Improve the Efficiency

- Combining two items at a time is a safe play
- If you know the exact cutoff of the likeness you want at every level, you can process multiple items at a time.
- Trade-off: you can no longer pick any arbitrary K .
- Let's do the next example on the board!



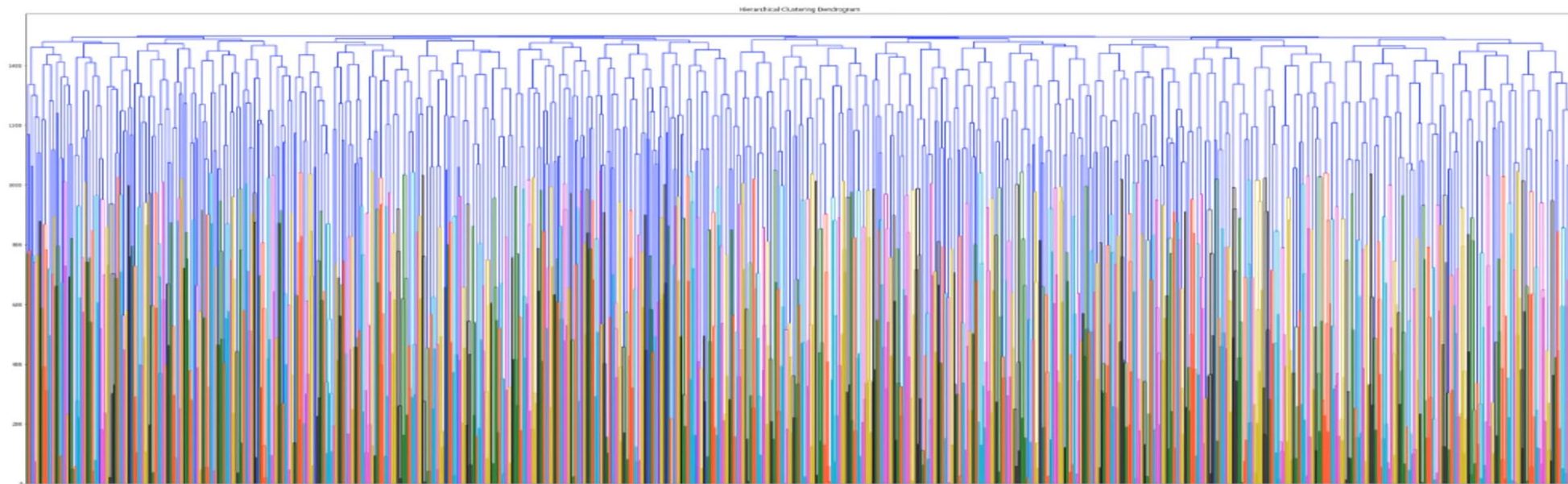
Divisive (Top-down)

- Start from 1 big cluster
- Divide into 2-clusters by some clustering method (k-means for example)
- After the method to divide the sub-clusters
- Repeat until each individual is its own cluster



Pros and Cons

- Hierarchical Clustering shows all the possible linkages between the clusters
- We understand the data much deeper
- No need to preset K
- Not scalable, computational very expensive
- Hard to interpret with merely 1000 samples
- Sensitive to outliers and noise



Time to Play

- Let's see how to use Hierarchical Clustering in **python**!

